

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain working principle of UV spectroscopy with its diagram.

Q.24 Discuss the infra-red FTIR spectroscopy with suitable diagram.

Q.25 Describe the Beer-Lambert's law.

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Roll No.

4th Sem / Instrumentation & Control

Subject : Analytical Instrumentation

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The first two elements constitute the_____.

- a) Processing module
- b) Characteristics module
- c) Display unit
- d) None of these

Q.2 Mass spectrometer is used to determine which of the following

- a) Composition in sample
- b) Relative mass of atoms
- c) Properties of sample
- d) None of these

Q.3 Which of the following are not used in liquid or HPLC chromatography

- a) Guard column b) Capillary column
c) Analytical column d) Separation column

Q.4 Range of PH scale is _____.

- a) 7 to 14 b) 0 to 14
c) 0 to 10 d) 7 to 10

Q.5 Advantages of PH meter is _____.

- a) Accurate reading b) User friendly
c) Robust d) All of these

Q.6 In the gas chromatography mobile phase is _____.

- a) A gas or a liquid b) Gas
c) Liquid d) None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Recording device is a part of UV spectroscopy. (True/False)

Q.8 Expand HPLC?

Q.9 AAS expands as Automatic Absorption spectroscopy. (True/False)

Q.10 Define oven.

Q.11 Electro-voltaic cells are driven by an external source of electrical energy. (True/False)

Q.12 Food analysis is application of gas chromatography. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Draw and briefly explain block diagram of analytical instruments.

Q.14 Discuss AAS.

Q.15 Explain the applications of FTIR.

Q.16 Write a short note on column and detectors for gas chromatography.

Q.17 Explain Liquid chromatography.

Q.18 Discuss PH scale.

Q.19 Describe the instrumentation used for HPLC.

Q.20 Explain PH meter.

Q.21 Discuss display unit as an element of an analytical instruments.

Q.22 Discuss applications of PH meter.